EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examinations 2018
General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME	

CIVICS GROUP	1	7	-			REGISTRATION NUMBER		

H1 Biology

8876/02

Paper 2 Structured Questions

14 September 2018

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on separate writing paper.
Additional Materials: Writing Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.
Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
7 or 8	
Total	60

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For Examiner's Use

Section A

Answer **all** the questions in this section.

1 The synthesis of collagen is shown in Fig. 1.1.

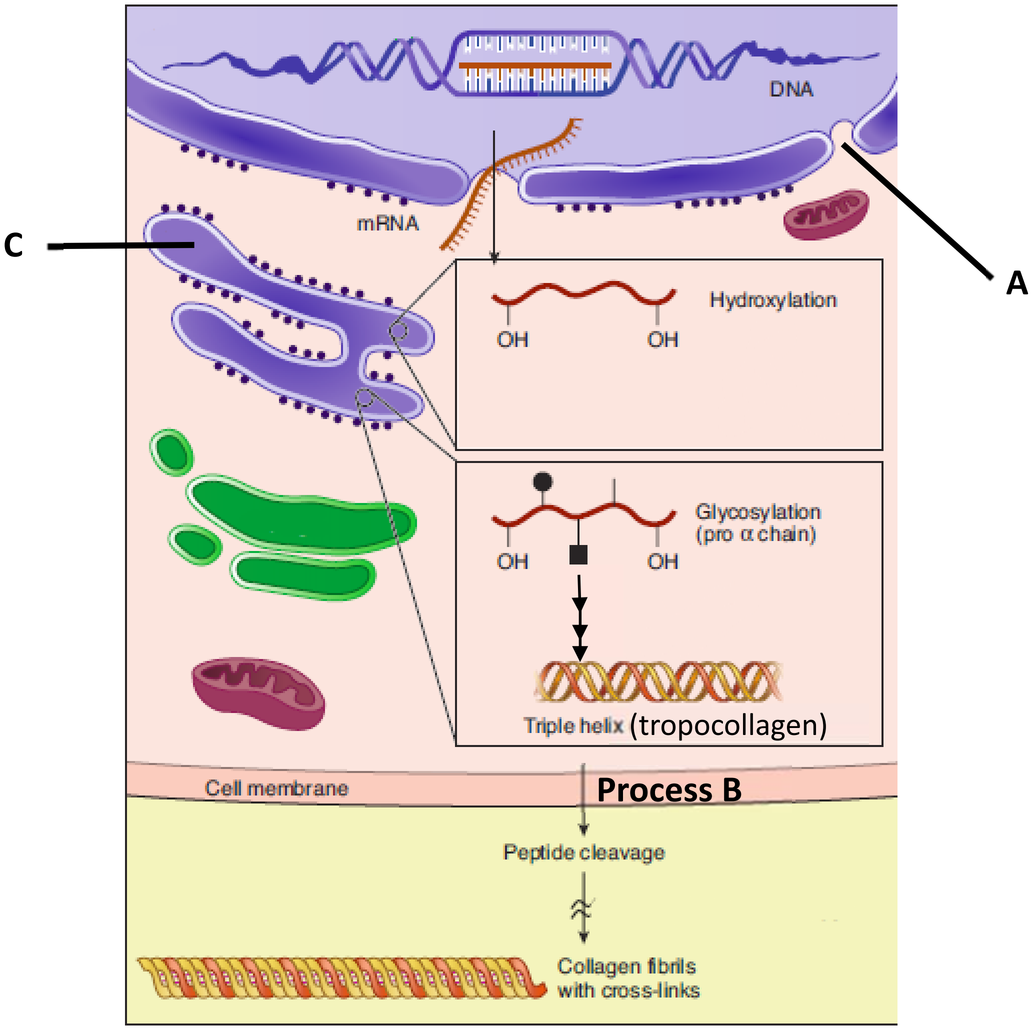

Fig. 1.1

(a) **(i)** Describe two important functions of structure **A** in the synthesis of collagen.

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....[2]

(ii) Tropocollagen leaves the cell to be assembled to form collagen fibrils via **Process B**. Outline **Process B**.

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Fig. 1.2 shows two electron micrographs. One of the electron micrograph shows part of a normal cell while the other electron micrograph shows part of a cancer cell. The white arrows point to an organelle within the cell. The appearance of this organelle in both cell types were visibly different. The cancer cell had a higher activity than the normal cell.

Organelle (indicated by white arrows) in a normal cell

Organelle (indicated by white arrows) in a cancer cell

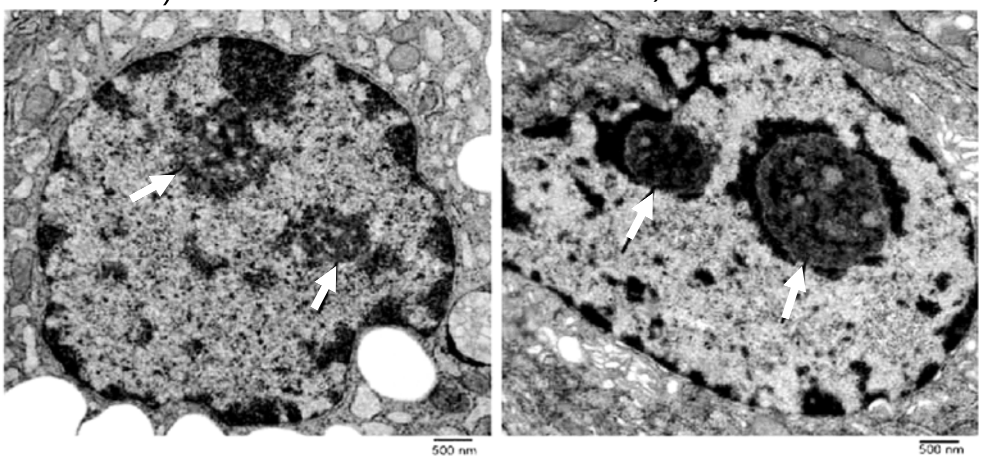

Fig. 1.2

(c) **(i)** Describe the visible difference between the organelle indicated by the white arrows in Fig. 1.2, between the cancer cell and the normal cell.

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(ii) Account for the higher activity of the cancer cell.

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[Total: 8]

2 Bone marrow contains stem cells that divide by mitosis to form blood cells. The fate of a stem cell was tracked and it was recorded that during the observed duration the stem cell divided asymmetrically each time.

Fig. 2.1 shows changes in the mass of DNA in a human stem cell from bone marrow during three cell cycles.

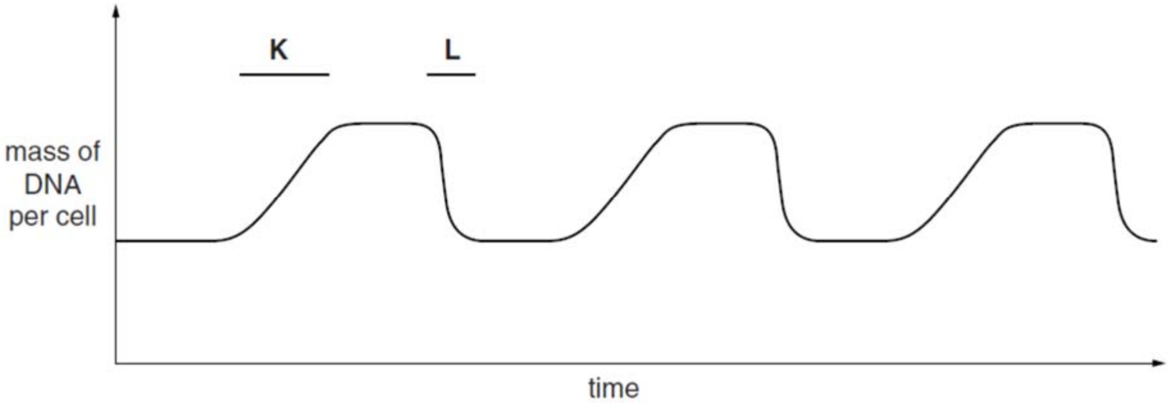

Fig. 2.1

(a) With reference to the information provided above,

(i) Describe what happens to bring about the changes in the mass of DNA per cell at time period **K** and at time period **L**.

K ……………………………………………………………………………………………

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L ……………………………………………………………………………………………

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(ii) State one function of these stem cells undergoing the above type of cell division.

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(iii) The process of meiosis is significant to natural selection in evolution.

Explain this significance.

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For Examiner's Use

The use of embryonic stem cells (ESCs) for stem cell therapy and research is controversial and considered by many people as unethical. Scientists have circumvented this issue through the use of induced pluripotent stem cells (iPSCs) as an alternative to ESCs.

Fig. 2.2 summarises the procedure for obtaining iPSCs and its use.

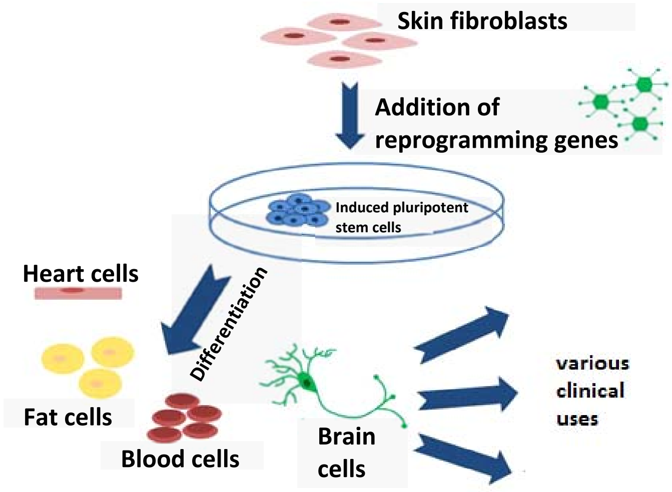

Fig. 2.2

(b) Explain why the use of iPSCs is preferred over ESCs.

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[Total: 7]

3 Fig. 3.1 shows part of a DNA molecule.

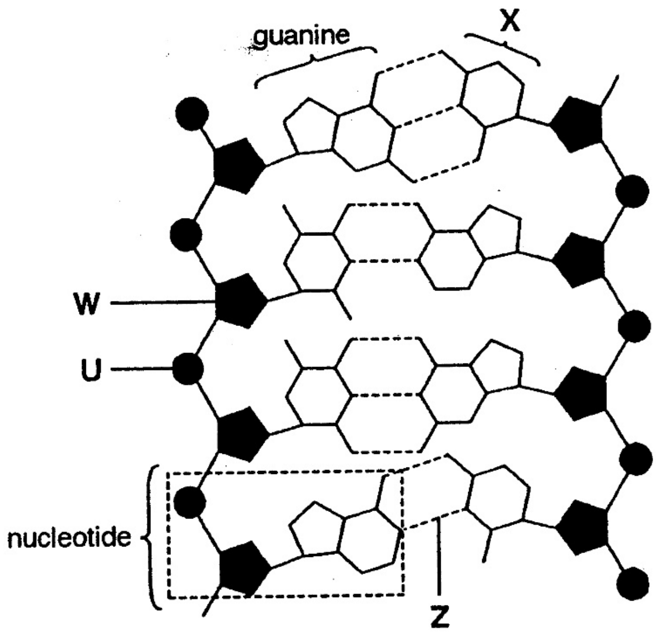

Fig. 3.1

(a) **(i)** Name **W** to **Z**. [1]

W ……………………………………………………

U ……………………………………………………

X ……………………………………………………

Z ……………………………………………………

(ii) Give **one** advantage of DNA having two strands.

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(b) Fig. 3.2 shows a linear chromosome undergoing the first round of DNA replication.

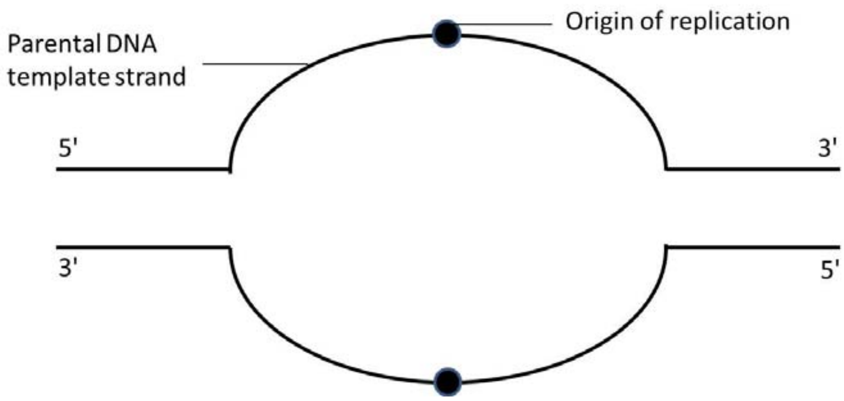

Fig. 3.2

(i) On Fig. 3.2, draw the direction of DNA synthesis for the leading strands using solid arrows (→) and for the lagging strands using dashed arrows (- - - >). [1]

(ii) State **two** ways DNA replication is different from transcription.

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(c) Fig. 3.3 is an electron micrograph showing the transcription of genes for ribosomal RNAs at adjacent positions along the chromosome in a eukaryotic cell.

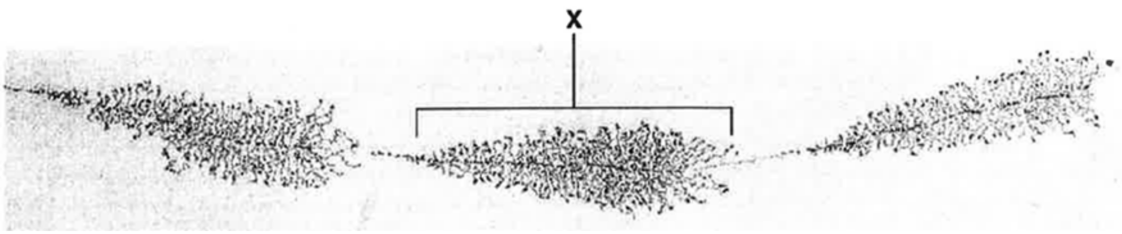

Fig. 3.3

Describe and explain the pattern of transcription visible on the part of the DNA coding for ribosomal RNA, labelled **X** in Fig. 3.3.

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[Total:7]

For Examiner's Use

4 Fig. 4.1 shows the pedigree of a family. Several members have the condition called albinism where there is a lack of normal pigment in the body.

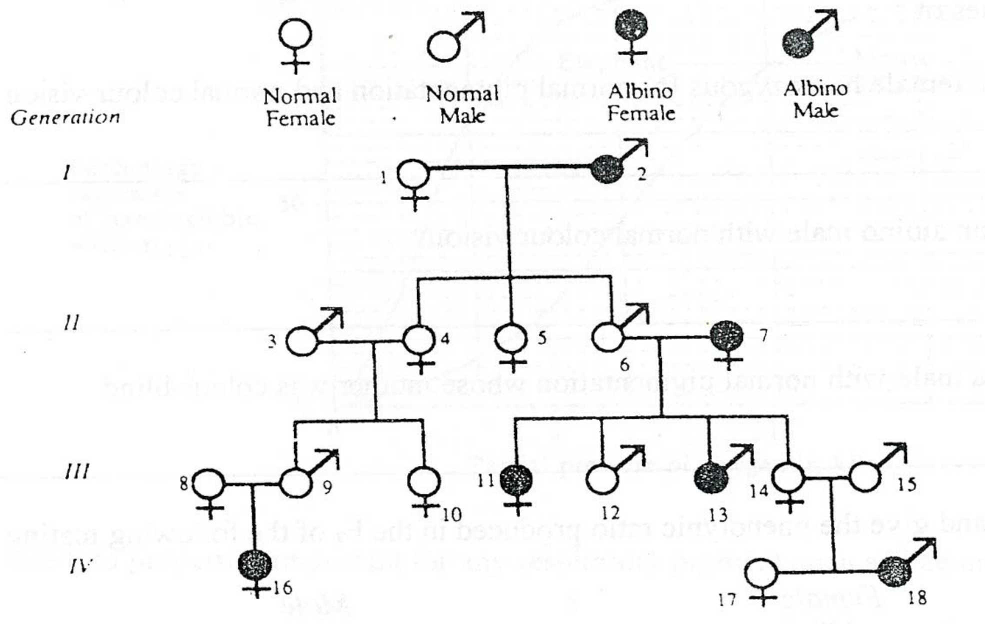

Fig. 4.1

(a) Skin colour is determined by an autosomal gene. Individuals who are not albino possess allele A which codes for an enzyme that synthesizes pigment. Allele a, on the other hand, codes for a non-functional enzyme.

What are all the possible genotypes of the following members of the family tree? [1]

individual 3 in generation II: ______________________

individual 14 in generation III: ______________________

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(b) The same family was also tested for colour blindness, a recessive sex-linked condition.

Individual 12 from generation III married a female with the same genotype as him at the A/a gene locus. Both of them are not colour blind, but one of their child is colour blind.

Use a genetic diagram to show the offspring phenotypic ratio of the mating between two such individuals.

[4]

[Total: 5]

For Examiner's Use

5 A student carried out an investigation on yeast respiration using the following setup shown in Fig. 5.1. She put 5 grams of yeast into a glucose solution and put a layer of oil on top of the mixture. This layer of oil is impermeable to oxygen but not to carbon dioxide. The total volume of gas was recorded every 10 minutes for 1 hour and the results are shown in Table 5.1.

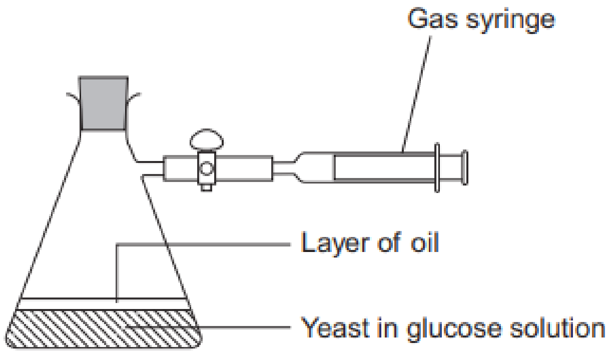

Fig. 5.1

Table 5.1

Time / minutes	Total volume of gas collected / cm^3
10	0.3
20	0.9
30	1.9
40	3.1
50	5.0
60	5.2

(a) **(i)** Calculate the rate of gas production in $cm^3\ g^{-1}\ min^{-1}$ during the first 40 minutes of this investigation. Show your working clearly. [1]

Answer = ... $cm^3\ g^{-1}\ min^{-1}$

For Examiner's Use

(ii) Suggest why the rate of gas production decreased between 50 and 60 minutes.

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(b) The Respiratory Quotient (RQ) is defined as the ratio of carbon dioxide produced to oxygen consumed per unit time by an organism and the formula is as follows:

$$RQ = \frac{\text{volume of } CO_2 \text{ produced}}{\text{volume of } O_2 \text{ consumed}} \text{ per unit time}$$

Given that the rate of carbon dioxide produced equals to the rate of oxygen used for aerobic respiration, suggest and explain the RQ value for the setup shown in Fig. 5.1.

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(c) To show that the presence of oil has an effect on yeast respiration, the student also created another similar setup as the one shown in Fig. 5.1. All variables were kept constant in this second setup except that no oil was used.

Explain if there is any difference in the amount of ATP produced between both setups.

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In another experiment, the student investigated the effect of temperature on the rate of oxygen consumption of the lizard, *Sauromalus hispidus*. The body temperature of a lizard varies with environmental temperature.

Several baby lizards were fitted with small, airtight masks that covered their heads. Air was supplied inside the mask through one tube, and collected through another. The differences between oxygen concentrations in the air supplied for inhalation and the exhaled air enabled the researchers to measure the rate of oxygen consumption of the lizards.

The rate of oxygen consumption of each baby lizard was measured at different temperatures ranging from 15 °C to 40 °C and the average measurements were computed. The results are shown in Fig. 5.2.

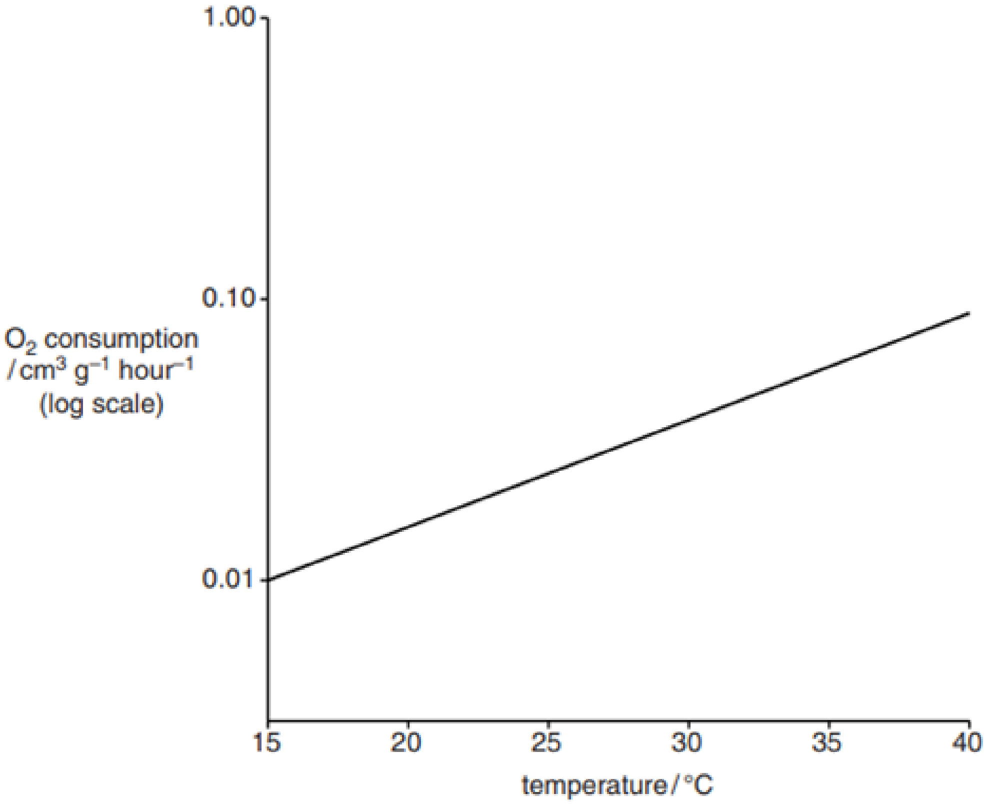

Fig. 5.2

(d) Using Fig. 5.2, suggest and explain how temperature influences the development of baby lizards.

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……[3]

[Total: 10]

6 The theory of continental drift was proposed over 90 years ago by Alfred Wegener and is now a widely accepted theory of how the continents were formed.

The supercontinent Gondwana is shown in Fig. 6.1. The land that was to form New Zealand began to break away from the supercontinent, Gondwana around 80 million years ago and broke away totally around 65 million years ago (Fig. 6.2). Ever since, New Zealand is now composed of North Island and South Island (Fig. 6.3).

The separation of New Zealand from Gondwana has led to some species like the New Zealand kiwi bird being distantly related to the rhea (South America), ostrich (Africa) and cassowary (Australia).

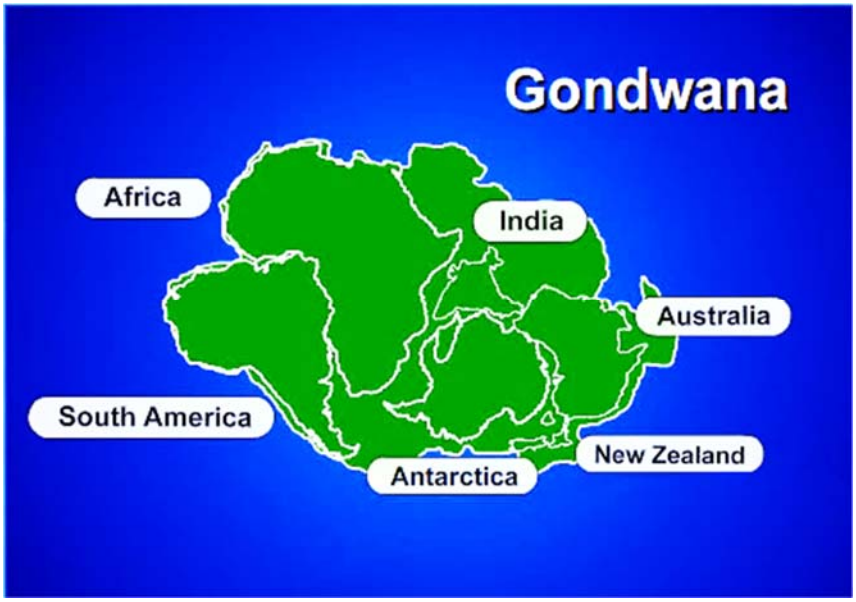

Fig. 6.1

Fig. 6.2

For Examiner's Use

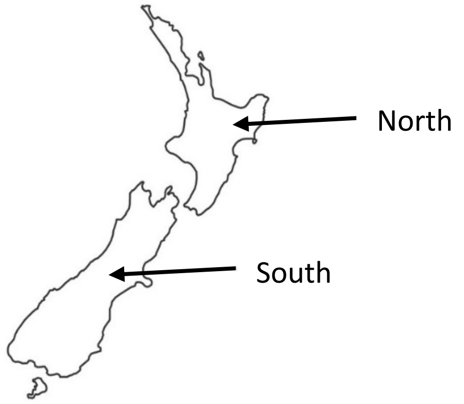

Fig. 6.3

For reasons that are not apparent, New Zealand was not inhabited by many mammals. Instead, its fauna became dominated by birds and insects. Birds became the predators, the scavengers, the herbivores and the insectivores. They lived everywhere from the highest mountains to the sea. Some have lost the ability to fly and became ground dwellers.

The kakapo, kea and kaka are parrots that are currently found in New Zealand (Fig. 6.4).

The kakapo (*Stringops habroptila*) is found only in New Zealand and is flightless. Highly endangered, they were relocated to several islands in New Zealand during the 1980s and 1990s.

The kea (*Nestor notabilis)* is the world's only alpine parrot, it can be found living near forested mountain habitats - Southern Alps on South Island. Kea are not found on North Island now.

Kaka (*Nestor meridionalis)* can be found on South Island and the sub-species of Kaka (*Nestor meridionalis septentrionalis*) is found on North Island. All Kaka live in the lowlands and mid-altitude regions.

kakapo

kea

kaka

Fig. 6.4

(a) Suggest a possible factor causing birds like the kakapo to become flightless early in their evolutionary history.

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………[1]

Fig. 6.5 shows the phylogenetic relationships between the kakapo, kea and kaka derived from mtDNA evidence, together with the geological changes that shape the landscape of New Zealand.

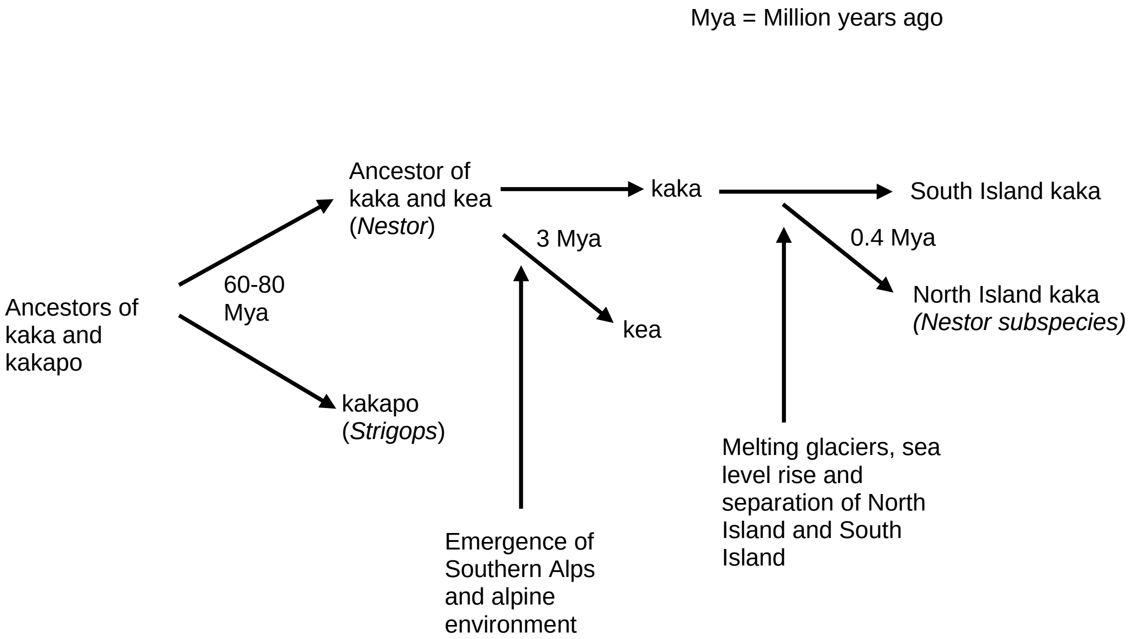

Fig. 6.5

(b) With reference to Fig. 6.5, explain the distribution and evolution of the kea.

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For Examiner's Use

(c) With reference to Fig. 6.5, suggest a possible reason for the emergence of a sub-species of kaka on North Island.

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……[2]

[Total:8]

For Examiner's Use

Section B

Answer either Q7 or Q8.

Write your answers for **Q7** or **Q8** on separate pieces of the answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 7

a) Discuss the importance of shapes fitting together in cells. [7]

b) Using named examples, describe the function of cycles in Biology. [8]

Question 8

a) Using examples, discuss the importance of movement in cellular processes. [7]

b) Polymers have different structures and functions. Describe how such structures are related to functions. [8]

- **END OF PAPER** -